# EXERCISE - O

## SINGLE CORRECT TYPE QUESTIONS

**1.** The value of the constant '$m$' and '$c$' for which $y = mx + c$ is a solution of the differential equation $D^2y - 3Dy - 4y = -4x$.

(A) is $m = -1; c = 3/4$ (B) is $m = 1; c = -3/4$

(C) no such real $m, c$ (D) is $m = 1;\ c = 3/4$

**MDE001**

**2.** Consider the two statements

Statement-1: $y = \sin kt$ satisfies the differential equation $y'' + 9y = 0$.

Statement-2: $y = e^{kt}$ satisfy the differential equation $y'' + y' - 6y = 0$

The value of $k$ for which both the statements are correct is

(A) – 3 (B) 0 (C) 2 (D) 3

**MDE002**

**3.** The $x$-intercept of the tangent to a curve is equal to the ordinate of the point of contact. The equation of the curve through the point (1, 1) is

(A) $ye^{\frac{x}{y}} = e$ (B) $xe^{\frac{x}{y}} = e$ (C) $xe^{\frac{y}{x}} = e$ (D) $ye^{\frac{y}{x}} = e$

**MDE006**

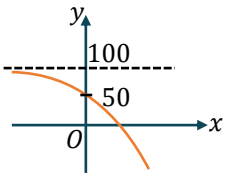
**4.** A function $f(x)$ satisfying $\int_0^1 f(tx)\,dt = nf(x)$, where $x > 0$, is

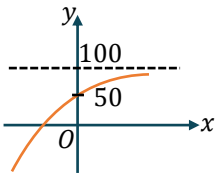
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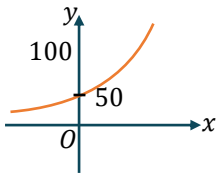
(A) $f(x) = c \cdot x^{\frac{1-n}{n}}$ (B) $f(x) = c \cdot x^{\frac{n}{n-1}}$ (C) $f(x) = c \cdot x^{\frac{1}{n}}$ (D) $f(x) = c \cdot x^{(1-n)}$

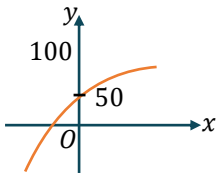
**MDE007**

**5.** Which one of the following curves represents the solution of the initial value problem $Dy = 100 - y$, where $y(0) = 50$

(A) 


(B) 


(C) 


(D) 


**MDE008**

**6.** The real value of $m$ for which the substitution, $y = u^m$ will transform the differential equation, $2x^4y\,\frac{dy}{dx} + y^4 = 4x^6$ into a homogeneous equation is :

(A) $m = 0$ (B) $m = 1$ (C) $m = 3/2$ (D) no value of $m$

**MDE009**

**7.** A function $y = f(x)$ satisfies $(x+1)\,.\,f'(x) - 2(x^2 + x)\,f(x) = \frac{e^{x^2}}{(x+1)}\ \forall\ x > -1$ ,

If $f(0) = 5$, then $f(x)$ is

(A) $\left(\frac{3x+5}{x+1}\right).e^{x^2}$ (B) $\left(\frac{6x+5}{x+1}\right).e^{x^2}$ (C) $\left(\frac{6x+5}{(x+1)^2}\right).e^{x^2}$ (D) $\left(\frac{5-6x}{x+1}\right).e^{x^2}$

**MDE011**



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**8.** If $\int_a^x t\,y(t)dt = x^2 + y(x)$ then $y$ as a function of $x$ is

(A) $y = 2 - (2 + a^2)\, e^{\frac{x^2+a^2}{2}}$ (B) $y = 1 - (2 + a^2)\, e^{\frac{x^2-a^2}{2}}$

(C) $y = 2 - (1 + a^2)\, e^{\frac{x^2-a^2}{2}}$ (D) none

**MDE012**

**9.** The population $p(t)$ at time $t$ of a certain mouse species satisfies the differential equation $\frac{dp(t)}{dt} = 0.5\,p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is :

(A) $\ell n18$ (B) $2\,\ell n18$ (C) $\ell n9$ (D) $\frac{1}{2}\ell n18$

**MDE013**

**10.** Let $I$ be the purchase value of an equipment and $V(t)$ be the value after it has been used for $t$ years. The value $V(t)$ depreciates at a rate given by differential equation = $-k(T - t)$, where $k > 0$ is a constant and $T$ is the total life in years of the equipment. Then the scrap value $V(T)$ of the equipment is :-

(A) $I - \frac{k(T-t)^2}{2}$ (B) $e^{-kT}$ (C) $T^2 - \frac{I}{k}$ (D) $I - \frac{kT^2}{2}$

**MDE014**

**11.** The equation to the orthogonal trajectories of the system of parabolas $y = ax^2$ is

(A) $\frac{x^2}{2} + y^2 = c$ (B) $x^2 + \frac{y^2}{2} = c$ (C) $\frac{x^2}{2} - y^2 = c$ (D) $x^2 - \frac{y^2}{2} = c$

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**MDE051**

**12.** The curve that passes through the point (2, 3), and has the property that the segment of any tangent to it lying between the coordinate axes is bisected by the point of contact, is given by :

(A) $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 2$ (B) $2y - 3x = 0$ (C) $y = \frac{6}{x}$ (D) $x^2 + y^2 = 13$

**MDE053**

## MULTIPLE CORRECT TYPE QUESTIONS

**13.** A curve $y = f(x)$ has the property that the perpendicular distance of the origin from the normal at any point $P$ of the curve is equal to the distance of the point $P$ from the $x$-axis. Then the differential equation of the curve

(A) is homogeneous.

(B) can be converted into linear differential equation with some suitable substitution.

(C) can corresponds to the family of circles touching the $x$-axis at the origin.

(D) can corresponds to the family of circles touching the $y$-axis at the origin.

**MDE057**



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**14.** If $y = f(x)$ is solution of the differential equation, $x^2 \frac{dy}{dx}.\cos\frac{1}{x} - y\sin\frac{1}{x} = -1$, where $y \to -1$ as $x \to \infty$, then

(A) $f(x) = \sin\frac{1}{x} - \cos\frac{1}{x}$

(B) $\lim_{x\to\infty}\left(-f(x)\right)^x = \frac{1}{e}$

(C) $\lim_{x\to\infty}\left(-f(x)\right)^x = e$

(D) $f'(x) + 2xf'(x) + x^2 f''(x) = -\frac{2}{x^2}\sin\frac{1}{x}$

**MDE058**

**15.** Correct statement is/are

(A) $f(x,y) = x^2 e^{\frac{x}{y}} + \frac{y^3}{x} + y^2 \ell n\left(\frac{y}{x}\right)$ = is a homogenous function of degree two.

(B) $f(x,y) = \frac{\sin y + x}{\sin 2y + x\cos y}$ is homogenous function of degree one.

(C) $x\sin\left(\frac{y}{x}\right)dy + \left(y\sin\frac{y}{x} - x\right)dx = 0$ is a homogenous differential equation.

(D) $f(x,y) = e^{\frac{y}{x}} + \tan\frac{y}{x}$ is homogenous function of degree zero.

**MDE059**

**16.** The solution of $x^2 y_1^2 + xy\, y^1 - 6y^2 = 0$ are (here $y_1 = dy/dx$)

(A) $y = Cx^2$

(B) $x^2 y = C$

(C) $\frac{1}{2}\ell ny = C + \log x$

(D) $x^3 y = C$

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**MDE060**

**17.** The solution of $\frac{dx}{dy} + y = ye^{(n-1)x}, (n \neq 1)$

(A) $\frac{1}{n-1}\ell n\left(\frac{e^{(n-1)x} - 1}{e^{(n-1)x}}\right) = \frac{y^2}{2} + C$

(B) $e^{(1-n)x} = 1 + ce^{(n-1)\frac{y^2}{2}}$

(C) $\ell n\left(1 + ce^{(n-1)\frac{y^2}{2}}\right) + nx + 1 = 0$

(D) $e^{(n-1)x} = ce^{(n-1)+\frac{(n-1)y^2}{2}} + 1$

**MDE063**

**18.** The differential equation for the family of curves $y = c\sin x$ can be given by

(A) $\left(\frac{dy}{dx}\right)^2 = y^2\cot^2 x$

(B) $\left(\frac{dy}{dx}\right)^2 - \left(\sec x\frac{dy}{dx}\right)^2 + y^2 = 0$

(C) $\left(\frac{dy}{dx}\right)^2 = \tan^2 x$

(D) $\frac{dy}{dx} = y\cot x$

**MDE064**

**19.** The solution of differential equation $\left(\frac{dy}{dx}\right)^2 - \frac{dy}{dx}(e^x + e^{-x}) + 1 = 0$ is

(A) $y = e^x + c$

(B) $y = -e^{-x} + c$

(C) $y = 2e^x + 3e^{-x} + c$

(D) $ye^x + 1 = ce^x$

**MDE066**



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## COMPREHENSION TYPE QUESTIONS

**Paragraph for Question No. 20 to 22**

Consider the polynomial $P(x) = (x - \cos36°)(x - \cos84°)(x - \cos156°)$

Differential equations are solved by reducing them to the exact differential of an expression in $x$ & $y$ i.e., they are reduced to the form $d(f(x, y)) = 0$

e.g. : $\frac{xdx + ydy}{\sqrt{x^2 + y^2}} = \frac{ydx - xdy}{x^2}$

$\Rightarrow \frac{1}{2}\frac{2xdx + 2ydy}{\sqrt{x^2 + y^2}} = -\frac{xdy - ydx}{x^2} \Rightarrow d\left(\sqrt{x^2 + y^2}\right) = -d\left(\frac{y}{x}\right)$

$\Rightarrow d\left(\sqrt{x^2 + y^2} + \frac{y}{x}\right) = 0$

$\therefore$ solution is $\sqrt{x^2 + y^2} + \frac{y}{x} = c$ .

Use the above method to answer the following question :

**20.** The general solution of $(2x^3 - xy^2)dx + (2y^3 - x^2y)dy = 0$ is

(A) $x^4 + x^2y^2 - y^4 = c$ (B) $x^4 - x^2y^2 + y^4 = c$

(C) $x^4 - x^2y^2 - y^4 = c$ (D) $x^4 + x^2y^2 + y^4 = c$

**MDE067**

**21.** General solution of the differential equation $\frac{xdy}{x^2 + y^2} + \left(1 - \frac{y}{x^2 + y^2}\right)dx = 0$ is

(A) $x + \tan^{-1}\left(\frac{y}{x}\right) = c$ (B) $x + \tan^{-1}\frac{x}{y} = c$

(C) $x - \tan^{-1}\left(\frac{y}{x}\right) = c$ (D) none of these

**MDE068**

**22.** General solution of the differential equation $e^y dx + (xe^y - 2y)\,dy = 0$ is

(A) $xe^y - y^2 = c$ (B) $ye^x - x^2 = c$ (C) $ye^y + x = c$ (D) $xe^y - 1 = cy^2$

**MDE069**

## MATRIX MATCH TYPE QUESTION

**23.** Match the following -

| | **List-I** | | **List- II** |
|---|---|---|---|
| (I) | Solution of $y - \frac{xdy}{dx} = y^2 + \frac{dy}{dx}$ is | (P) | $xy^2 = 2y^5 + c$ |
| (II) | Solution of $(2x - 10y^3)\frac{dy}{dx} + y = 0$ is | (Q) | $\sec y = x + 1 + ce^x$ |
| (III) | Solution of $\sec^2 y\,dy + \tan y\,dx = dx$ is | (R) | $(x + 1)(1 - y) = cy$ |
| (IV) | Solution of $\sin y \frac{dy}{dx} = \cos y(1 - x\cos y)$ is | (S) | $\tan y = 1 + ce^{-x}$ |

(A) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ S (B) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ Q

(C) I $\rightarrow$ P; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ Q (D) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ Q

**MDE070**

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## EXERCISE - S

**1.** If $y = y(x)$ is the solution of the differential equation, $x\frac{dy}{dx}+2y=x^2$ satisfying $y(1) = 1$, if $y\left(\frac{1}{2}\right)=k$, then find the value of $16k$ :

**MDE016**

**2.** Let $f$ be a differentiable function such that $f'(x)=7-\frac{3}{4}\frac{f(x)}{x},(x>0)$ and $f(1) \neq 4$. Then the value of $\lim\limits_{x\to 0^+} x f\left(\frac{1}{x}\right)$ is :

**MDE017**

**3.** Let $y = y(x)$ be the solution of the differential equation, $\left(x^2+1\right)^2\frac{dy}{dx}+2x\left(x^2+1\right)y=1$ such that $y(0) = 0$. If $\sqrt{a}y(1)=\frac{\pi}{32}$, then the value of '$32a$' is :

**MDE018**

**4.** If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} = (\tan x - y)\sec^2 x,\ x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, such that $y(0) = 0$, then $y\left(-\frac{\pi}{4}\right)$ is equal to $(e - k)$, find $k$

**MDE019**

**5.** Let $y = y(x)$ be the solution of the differential equation, $\frac{2+\sin x}{y+1}\cdot\frac{dy}{dx}=-\cos x, y>0, y(0)=1$ . If $y(\pi) = a$ and $\frac{dy}{dx}$ at $x = \pi$ is $b$, then the value of $(a + b)$ is

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**MDE020**

**6.** If $y = y(x)$ is the solution of the differential equation $\frac{5+e^x}{2+y}\cdot\frac{dy}{dx}+e^x=0$ satisfying $y(0) = 1$, then the value of $|y(\log_e 13)|$ is

**MDE021**

**7.** If $y=\left(\frac{2}{\pi}x-1\right)\text{cosec}x$ is the solution of the differential equation, $\frac{dy}{dx}+p(x)y=\frac{2}{\pi}\text{cosec}x, 0<x<\frac{\pi}{2}$, then find the value of $P\left(\frac{\pi}{4}\right)$.

**MDE022**

**8.** If $y(x)$ satisfies the equation $y'(x) = y(x) + \int_0^1 y\,dx$ & $y(0) = 1$ then value of $y\left(\ell n\frac{11-3e}{2}\right)$

**MDE023**

**9.** If solution of the differential equation $\frac{dy}{dx}=\frac{1}{x\cos y+\sin 2y}$ is $x = ce^{\sin y} - k(1 + \sin y)$, then $k =$

**MDE024**

**10.** If the equation of curve passing through (3, 4) and satisfying the differential equation $y\left(\frac{dy}{dx}\right)^2+(x-y)\frac{dy}{dx}-x=0$ is $Ax + By + 2 = 0$ then value of $(A - B)$ is

**MDE025**



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# EXERCISE - JEE (Main) PYQ

**1.** Let the population of rabbits surviving at a time $t$ be governed by the differential equation $\frac{dp(t)}{dt}=\frac{1}{2}p(t)-200$. If $p(0)=100$, then $p(t)$ equals: **[JEE (Main) 2014]**

(1) $400-300\,e^{\frac{t}{2}}$ (2) $300-200\,e^{-\frac{t}{2}}$ (3) $600-500\,e^{\frac{t}{2}}$ (4) $400-300\,e^{-\frac{t}{2}}$

**MDE028**

**2.** Let $y(x)$ be the solution of the differential equation $(x\log x)\frac{dy}{dx}+y=2x\log x,(x\geq 1)$.

Then $y(e)$ is equal to: **[JEE (Main) 2015]**

(1) 2 (2) $2e$ (3) $e$ (4) 0

**MDE029**

**3.** If a curve $y=f(x)$ passes through the point (1, –1) and satisfies the differential equation, $y(1+xy)\,dx=x\,dy$, then $f\left(-\frac{1}{2}\right)$ is equal to : **[JEE (Main) 2016]**

(1) $\frac{4}{5}$ (2) $-\frac{2}{5}$ (3) $-\frac{4}{5}$ (4) $\frac{2}{5}$

**MDE030**

**4.** Let $y=y(x)$ be the solution of the differential equation $\sin x\frac{dy}{dx}+y\cos x=4x, x\in(0,\pi)$. If $y\left(\frac{\pi}{2}\right)=0$, then $y\left(\frac{\pi}{6}\right)$ is equal to: **[JEE (Main) 2018]**

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(1) $\frac{-8}{9\sqrt{3}}=0\,\pi^2$ (2) $-\frac{8}{9}\pi^2$ (3) $-\frac{4}{9}\pi^2$ (4) $\frac{4}{9\sqrt{3}}\pi^2$

**MDE032**

**5.** Let $f:[0,1]\to R$ be such that $f(xy)=f(x).f(y)$ for all $x,y,\in[0,1]$, and $f(0)\neq 0$. If $y=y(x)$ satisfies the differential equation, $\frac{dy}{dx}=f(x)$ with $y(0)=1$, then $y\left(\frac{1}{4}\right)+y\left(\frac{3}{4}\right)$ is equal to **[JEE (Main) 2019]**

(1) 4 (2) 3 (3) 5 (4) 2

**MDE033**

**6.** If $\frac{dy}{dx}+\frac{3}{\cos^2 x}y=\frac{1}{\cos^2 x}, x\in\left(\frac{-\pi}{3},\frac{\pi}{3}\right)$ and $y\left(\frac{\pi}{4}\right)=\frac{4}{3}$, then $y\left(-\frac{\pi}{4}\right)$ equals : **[JEE (Main) 2019]**

(1) $\frac{1}{3}+e^6$ (2) $\frac{1}{3}$ (3) $-\frac{4}{3}$ (4) $\frac{1}{3}+e^3$

**MDE034**

**7.** The curve amongst the family of curves, represented by the differential equation, $(x^2-y^2)dx+2xy\,dy=0$ which passes through (1,1) is : **[JEE (Main) 2019]**

(1) A circle with centre on the $y$-axis
(2) A circle with centre on the $x$-axis
(3) An ellipse with major axis along the $y$-axis
(4) A hyperbola with transverse axis along the $x$-axis

**MDE035**



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**8.** If a curve passes through the point (1, –2) and has slope of the tangent at any point $(x, y)$ on it as $\frac{x^2-2y}{x}$, then the curve also passes through the point : **[JEE (Main) 2019]**

(1) $\left(-\sqrt{2},1\right)$ (2) $\left(\sqrt{3},0\right)$ (3) (–1, 2) (4) (3, 0)

**MDE037**

**9.** Let $y = y(x)$ be the solution curve of the differential equation, $(y^2 - x)\frac{dy}{dx} = 1$, satisfying $y(0) = 1$. This curve intersects the $x$-axis at a point whose abscissa is : **[JEE (Main) 2020]**

(1) $2 + e$ (2) 2 (3) $2 - e$ (4) $-e$

**MDE038**

**10.** If for $x \geq 0$, $y = y(x)$ is the solution of the differential equation $(x + 1)dy = ((x + 1)^2 + y - 3)dx$, $y(2) = 0$, then $y(3)$ is equal to . **[JEE (Main) 2020]**

**MDE039**

**11.** If a curve $y = f(x)$ passes through the point (1, 2) and satisfies $x\frac{dy}{dx} + y = bx^4$, then for what value of $b$, $\int_1^2 f(x)dx = \frac{62}{5}$ ? **[JEE (Main) 2021]**

(1) 5 (2) 10 (3) $\frac{62}{5}$ (4) $\frac{31}{5}$

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**MDE040**

**12.** If a curve passes through the origin and the slope of the tangent to it at any point $(x, y)$ is $\frac{x^2-4x+y+8}{x-2}$, then this curve also passes through the point : **[JEE (Main) 2021]**

(1) (5, 4) (2) (4, 5) (3) (4, 4) (4) (5, 5)

**MDE042**

**13.** A particle is moving in the $xy$-plane along a curve $C$ passing through the point (3, 3). The tangent to the curve $C$ at the point $P$ meets the $x$-axis at $Q$. If the $y$-axis bisects the segment $PQ$, then $C$ is a parabola with **[JEE (Main) 2022]**

(1) length of latus rectum 3 (2) length of latus rectum 6

(3) focus $\left(\frac{4}{3},0\right)$ (4) focus $\left(0,\frac{3}{4}\right)$

**MDE043**

**14.** The slope of normal at any point $(x, y), x > 0, y > 0$ on the curve $y = y(x)$ is given by $\frac{x^2}{xy - x^2y^2 - 1}$. If the curve passes through the point (1, 1), then $e.y(e)$ is equal to **[JEE (Main) 2022]**

(1) $\frac{1-\tan(1)}{1+\tan(1)}$ (2) $\tan(1)$ (3) 1 (4) $\frac{1+\tan(1)}{1-\tan(1)}$

**MDE044**



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**15.** Let $y = y(x)$ be a solution of the differential equation $(x\cos x)dy + (xy\sin x + y\cos x - l)dx = 0$, $0 < x < \frac{\pi}{2}$. If $\frac{\pi}{3}y\left(\frac{\pi}{3}\right) = \sqrt{3}$, then $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right) + 2y'\left(\frac{\pi}{6}\right)\right|$ is equal to _____ **[JEE (Main) 2023]**

**MDE047**

**16.** Let the solution curve $x = x(y)$, $0 < y < \frac{\pi}{2}$, of the differential equation

$\left(\log_e(\cos y)\right)^2 \cos y dx - \left(1 + 3x\log_e(\cos y)\right)\sin y dy = 0$

satisfy $x\left(\frac{\pi}{3}\right) = \frac{1}{2\log_e 2}$. If $x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_e n}$, where $m$ and $n$ are co-prime, then mn is equal to

**[JEE (Main) 2023]**

**MDE048**

**17.** Let $f$ be a differentiable function such that $x^2 f(x) - x = 4\int_0^x t f(t)dt$, $f(1) = \frac{2}{3}$.

Then $18 f(3)$ is equal to **[JEE (Main) 2023]**

(1) 160 (2) 210 (3) 180 (4) 150

**MDE050**

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# EXERCISE - JEE (Advanced) PYQ

**1.** Let $f:\left[\frac{1}{2},1\right]\to R$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x)<2f(x)$ and $f\left(\frac{1}{2}\right)=1$. Then the value of $\int_{1/2}^{1} f(x)dx$ lies in the interval **[JEE (Advanced) 2013]**

(A) $(2e-1, 2e)$ (B) $(e-1, 2e-1)$

(C) $\left(\frac{e-1}{2}, e-1\right)$ (D) $\left(0, \frac{e-1}{2}\right)$

**MDE071**

**2.** A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point $(x, y)$ be $\frac{y}{x}+\sec\left(\frac{y}{x}\right)$, $x>0$. Then the equation of the curve is **[JEE (Advanced) 2013]**

(A) $\sin\left(\frac{y}{x}\right)=\log x+\frac{1}{2}$ (B) $\text{cosec}\left(\frac{y}{x}\right)=\log x+2$

(C) $\sec\left(\frac{2y}{x}\right)=\log x+2$ (D) $\cos\left(\frac{2y}{x}\right)=\log x+\frac{1}{2}$

**MDE072**

**Paragraph for Question 3 and 4**

Let $f:[0,1]\to$ IR (the set of all real numbers) be a function. Suppose the function $f$ is twice differentiable, $f(0)=f(1)=0$ and satisfies $f''(x)-2f'(x)+f(x)\geq e^x, x\in[0,1]$.

**3.** If the function $e^{-x}f(x)$ assumes its minimum in the interval $[0,1]$ at $x=\frac{1}{4}$, which of the following is true? **[JEE (Advanced) 2013]**

(A) $f'(x)<f(x), \frac{1}{4}<x<\frac{3}{4}$ (B) $f'(x)>f(x), 0<x<\frac{1}{4}$

(C) $f'(x)<f(x), 0<x<\frac{1}{4}$ (D) $f'(x)<f(x), \frac{3}{4}<x<1$

**MDE073**

**4.** Which of the following is true for $0<x<1$? **[JEE (Advanced) 2013]**

(A) $0<f(x)<\infty$ (B) $-\frac{1}{2}<f(x)<\frac{1}{2}$ (C) $-\frac{1}{4}<f(x)<1$ (D) $-\infty<f(x)<0$

**MDE074**

**5.** The function $y=f(x)$ is the solution of the differential equation $\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{x^4+2x}{\sqrt{1-x^2}}$ in (-1, 1) satisfying $f(0)=0$. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is **[JEE (Advanced) 2014]**

(A) $\frac{\pi}{3}-\frac{\sqrt{3}}{2}$ (B) $\frac{\pi}{3}-\frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6}-\frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6}-\frac{\sqrt{3}}{2}$

**MDE075**



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**6.** Let $y(x)$ be a solution of the differential equation $(1+e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is(are) true ? **[JEE (Advanced) 2015]**

(A) $y(-4) = 0$
(B) $y(-2) = 0$
(C) $y(x)$ has a critical point in the interval (–1,0)
(D) $y(x)$ has no critical point in the interval (–1,0)

**MDE076**

**7.** Consider the family of all circles whose centres lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where $P, Q$ are functions of $x, y$ and $y'$ (here $y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}$), then which of the following statements is (are) true ?

**[JEE (Advanced) 2015]**

(A) $P = y + x$ (B) $P = y - x$
(C) $P + Q = 1 - x + y + y' + (y')^2$ (D) $P - Q = x + y - y' - (y')^2$

**MDE077**

**8.** A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4)\frac{dy}{dx} - y^2 = 0, x > 0$, passes through the point (1, 3). The solution curve- **[JEE (Advanced) 2016]**

(A) intersects $y = x + 2$ exactly at one point (B) intersects $y = x + 2$ exactly at two points
(C) intersects $y = (x + 2)^2$ (D) does NOT intersect $y = (x + 3)^2$

**MDE078**

**9.** Let $f:(0,\infty) \to \mathbb{R}$ be a differentiable function such that $f'(x) = 2 - \frac{f(x)}{x}$ for all $x \in (0,\infty)$ and $f(1) \neq 1$. Then **[JEE (Advanced) 2016]**

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(A) $\lim\limits_{x\to 0^+} f'\left(\frac{1}{x}\right) = 1$ (B) $\lim\limits_{x\to 0^+} xf\left(\frac{1}{x}\right) = 2$
(C) $\lim\limits_{x\to 0^+} x^2 f'(x) = 0$ (D) $|f(x)| \leq 2$ for all $x \in (0,2)$

**MDE079**

**10.** If $y = y(x)$ satisfies the differential equation $8\sqrt{x}\left(\sqrt{9+\sqrt{x}}\right)dy = \left(\sqrt{4+\sqrt{9+\sqrt{x}}}\right)^{-1} dx$, $x > 0$ and $y(0) = \sqrt{7}$, then $y(256) =$ **[JEE (Advanced) 2017]**

(A) 80 (B) 3 (C) 16 (D) 9

**MDE080**

**11.** If $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function such that $f'(x) > 2f(x)$ for all $x \in \mathbb{R}$, and $f(0) = 1$, then **[JEE (Advanced) 2017]**

(A) $f(x) > e^{2x}$ in $(0,\infty)$ (B) $f(x)$ is decreasing in $(0,\infty)$
(C) $f(x)$ is increasing in $(0,\infty)$ (D) $f'(x) < e^{2x}$ in $(0,\infty)$

**MDE081**

**12.** Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be two non-constant differentiable functions. If $f'(x) = \left(e^{(f(x)-g(x))}\right)g'(x)$ for all $x \in \mathbb{R}$, and $f(1) = g(2) = 1$, then which of the following statement(s) is (are) TRUE ? **[JEE (Advanced) 2018]**

(A) $f(2) < 1 - \log_e 2$ (B) $f(2) > 1 - \log_e 2$ (C) $g(1) > 1 - \log_e 2$ (D) $g(1) < 1 - \log_e 2$

**MDE082**



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**13.** Let $f:(0,\pi)\to\mathbb{R}$ be a twice differentiable function such that $\lim\limits_{t\to x}\dfrac{f(x)\sin t-f(t)\sin x}{t-x}=\sin^2 x$ for all $x\in(0,\pi)$. If $f\left(\dfrac{\pi}{6}\right)=-\dfrac{\pi}{12}$, then which of the following statement(s) is (are) TRUE ?

**[JEE (Advanced) 2018]**

(A) $f\left(\dfrac{\pi}{4}\right)=\dfrac{\pi}{4\sqrt{2}}$ (B) for all $x\in(0,\pi)$

(C) There exists $\alpha\in(0,\pi)$ such that $f'(\alpha)=0$ (D) $f''\left(\dfrac{\pi}{2}\right)+f\left(\dfrac{\pi}{2}\right)=0$

**MDE083**

**14.** Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function with $f(0)=0$. If $y=f(x)$ satisfies the differential equation $\dfrac{dy}{dx}=(2+5y)(5y-2)$, then the value of $\lim\limits_{x\to-\infty}f(x)$ is ____. **[JEE (Advanced) 2018]**

**MDE084**

**15.** Let $\Gamma$ denote a curve $y=y(x)$ which is in the first quadrant and let the point $(1,0)$ lie on it. Let the tangent to $\Gamma$ at a point $P$ intersect the y-axis at $Y_P$. If $PY_P$ has length 1 for each point $P$ on $\Gamma$, then which of the following options is/are correct ? **[JEE (Advanced) 2019]**

(A) $y=\log_e\left(\dfrac{1+\sqrt{1-x^2}}{x}\right)-\sqrt{1-x^2}$ (B) $xy'-\sqrt{1-x^2}=0$

(C) $y=-\log_e\left(\dfrac{1+\sqrt{1-x^2}}{x}\right)+\sqrt{1-x^2}$ (D) $xy'+\sqrt{1-x^2}=0$

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**MDE085**

**16.** Let $b$ be a nonzero real number. Suppose $f:\mathbb{R}\to\mathbb{R}$ is a differentiable function such that $f(0)=1$. If the derivative $f'$ of $f$ satisfies the equation $f'(x)=\dfrac{f(x)}{b^2+x^2}$ for all $x\in\mathbb{R}$, then which of the following statements is/are TRUE ? **[JEE (Advanced) 2020]**

(A) If $b>0$, then $f$ is an increasing function (B) If $b<0$, then $f$ is a decreasing function

(C) $f(x)\,f(-x)=1$ for all $x\in\mathbb{R}$ (D) $f(x)-f(-x)=0$ for all $x\in\mathbb{R}$

**MDE086**

**17.** For any real numbers $\alpha$ and $\beta$, let $y_{\alpha,\beta}(x)$, $x\in\mathbb{R}$, be the solution of the differential equation $\dfrac{dy}{dx}+\alpha y=xe^{\beta x}, y(1)=1$ **[JEE (Advanced) 2021]**

Let $S=\{y_{\alpha,\beta}(x):\alpha,\beta\in\mathbb{R}\}$. Then which of the following functions belong(s) to the set $S$?

(A) $f(x)=\dfrac{x^2}{2}e^{-x}+\left(e-\dfrac{1}{2}\right)e^{-x}$ (B) $f(x)=-\dfrac{x^2}{2}e^{-x}+\left(e+\dfrac{1}{2}\right)e^{-x}$

(C) $f(x)=\dfrac{e^x}{2}\left(x-\dfrac{1}{2}\right)+\left(e-\dfrac{e^2}{4}\right)e^{-x}$ (D) $f(x)=\dfrac{e^x}{2}\left(\dfrac{1}{2}-x\right)+\left(e+\dfrac{e^2}{4}\right)e^{-x}$

**MDE087**



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**18.** If $y(x)$ is the solution of the differential equation $xdy-(y^2-4y)dx=0$ for $x>0, y(1)=2$, and the slope of the curve $y=y(x)$ is never zero, then the value of $10y\left(\sqrt{2}\right)$ is _____.

**[JEE (Advanced) 2022]**

**MDE088**

**19.** For $x\in\mathbb{R}$, let the function $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx}+12y=\cos\left(\frac{\pi}{12}x\right), y(0)=0.$$

Then, which of the following statements is/are TRUE ? **[JEE (Advanced) 2022]**

(A) $y(x)$ is an increasing function

(B) $y(x)$ is a decreasing function

(C) There exists a real number β such that the line $y=\beta$ intersects the curve $y=y(x)$ at infinitely many points

(D) $y(x)$ is a periodic function

**MDE089**

**20.** Let $f:[(1,\infty)\to\mathbb{R}$ be a differentiable function such that $f(1)=\frac{1}{3}$ and

$3\int_1^x f(t)dt=xf(x)-\frac{x^3}{3}, x\in[1,\infty)$. Let $e$ denote the base of the natural logarithm. Then the value of $f(e)$ is **[JEE (Advanced) 2023]**

(A) $\frac{e^2+4}{3}$ (B) $\frac{\log_e 4+e}{3}$ (C) $\frac{4e^2}{3}$ (D) $\frac{e^2-4}{3}$

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**MDE090**

**21.** Let $S$ be the set of all twice differentiable functions $f$ from $\mathbb{R}$ to $\mathbb{R}$ such that $\frac{d^2f}{dx^2}(x)>0$ for all $x\in(-1,1)$. For $f\in S$, let $X_f$ be the number of points $x\in(-1,1)$ for which $f(x)=x$. Then which of the following statements is(are) true? **[JEE (Advanced) 2023]**

(A) There exists a function $f\in S$ such that $X_f=0$

(B) For every function $f\in S$, we have $X_f\le 2$

(C) There exists a function $f\in S$ such that $X_f=2$

(D) There does **NOT** exist any function $f$ in $S$ such that $X_f=1$

**MDE091**

**22.** For $x\in\mathbb{R}$, let $y(x)$ be a solution of the differential equation $(x^2-5)\frac{dy}{dx}-2xy=-2x(x^2-5)^2$ such that $y(2)=7$. Then the maximum value of the function $y(x)$ is **[JEE (Advanced) 2023]**

**MDE092**



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# JEE (Main) Practice Paper

This paper is for yourself practice and assessment the discussion of this paper is optional though you can see PDF solutions or video solutions or solutions in hardcopy whichever is provided.

## SECTION–A

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (1), (2), (3) and (4). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:

*Full Marks* : +4, if only the bubble corresponding to the correct option is darkened.
*Zero Marks* : 0, if none of the bubbles is darkened.
*Negative Marks* : –1 in all other cases.

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**1.** The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$, where $c_1$ and $c_2$ are arbitrary constants, is :-

(1) $yy'' = y'$ (2) $yy'' = (y')^2$ (3) $y' = y^2$ (4) $y'' = y'y$

**MDE093**

**2.** Let $I$ be the purchase value of an equipment and $V(t)$ be the value after it has been used for $t$ years. The value $V(t)$ depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T-t)$, where $k > 0$ is a constant and $T$ is the total life in years of the equipment. Then the scrap value $V(T)$ of the equipment is :-

(1) $I - \frac{k(T-t)^2}{2}$ (2) $e^{-kT}$ (3) $T^2 - \frac{I}{k}$ (4) $I - \frac{kT^2}{2}$

**MDE094**

**3.** The curve that passes through the point (2, 3), and has the property that the segment of any tangent to it lying between the coordinate axes is bisected by the point of contact, is given by:

(1) $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 2$ (2) $2y - 3x = 0$ (3) $y = \frac{6}{x}$ (4) $x^2 + y^2 = 13$

**MDE095**

**4.** Consider the differential equation $y^2 dx + \left(x - \frac{1}{y}\right)dy = 0$. It $y(1) = 1$, then $x$ is given by :

(1) $1 - \frac{1}{y} + \frac{\frac{1}{e^y}}{e}$ (2) $4 - \frac{2}{y} - \frac{\frac{1}{e^y}}{e}$ (3) $3 - \frac{1}{y} + \frac{\frac{1}{e^y}}{e}$ (4) $1 + \frac{1}{y} - \frac{\frac{1}{e^y}}{e}$

**MDE096**

**5.** The population $p(t)$ at time $t$ of a certain mouse species satisfies the differential equation $\frac{dp(t)}{dt} = 0.5\, p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is:

(1) $ln 18$ (2) $2\, ln 18$ (3) $ln 9$ (4) $\frac{1}{2} ln 18$

**MDE097**



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**6.** A spherical balloon is filled with $4500\pi$ cubic meters of helium gas. If a leak in the balloon causes the gas to escape at the rate of $72\pi$ cubic meters per minute, then the rate (in meters per minute) at which the radius of the balloon decreases 49 minutes after the leakage began is :

(1) 9/2 (2) 9/7 (3) 7/9 (4) 2/9

**MDE098**

**7.** The differential equation for all the straight lines which are at a unit distance from the origin is

(1) $\left(y-x\frac{dy}{dx}\right)^2 = 1-\left(\frac{dy}{dx}\right)^2$ (2) $\left(y+x\frac{dy}{dx}\right)^2 = 1+\left(\frac{dy}{dx}\right)^2$

(3) $\left(y-x\frac{dy}{dx}\right)^2 = 1+\left(\frac{dy}{dx}\right)^2$ (4) $\left(y+x\frac{dy}{dx}\right)^2 = 1-\left(\frac{dy}{dx}\right)^2$

**MDE099**

**8.** If the differential equation representing the family of all circles touching x-axis at the origin is $(x^2-y^2)\frac{dy}{dx} = g(x)\,y$, then $g(x)$ equals

(1) $2x^2$ (2) $\frac{1}{2}x^2$ (3) $\frac{1}{2}x$ (4) $2x$

**MDE100**

**9.** If $\phi(x) = \phi'(x)$ and $\phi(1) = 2$, then $\phi(3)$ equals

(1) $e^2$ (2) $2e^2$ (3) $3e^2$ (4) $2e^3$

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**MDE101**

**10.** If $\frac{dy}{dx} = 1+x+y+xy$ and $y(-1) = 0$, then function $y$ is

(1) $e^{(1-x)^2/2}$ (2) $e^{(1+x)^2/2}-1$ (3) $\ell n(1+x)-1$ (4) $1+x$

**MDE102**

**11.** The solution of the differential equation $\frac{dy}{dx}-ky = 0, y(0) = 1$, approaches zero when $x\to\infty$, if

(1) $k = 0$ (2) $k > 0$ (3) $k < 0$ (4) $k \geq 0$

**MDE103**

**12.** The general solution of the differential equation $ydy + \sqrt{1+y^2}\,dx = 0$ represents a family of :

(1) circles (2) ellipses other than circles

(3) hyperbolas (4) parabolas

**MDE104**

**13.** The conic whose differential equation is $(1+y^2)dx - xydy = 0$ and which passess through (1, 0) is

(1) $x^2-y^2 = 1$ (2) $2x^2-3y^2 = 1$ (3) $x^2+y^2 = 1$ (4) $x^2-y^2 = 2$

**MDE105**



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**14.** The general solution of the differential equation $\frac{dy}{dx} + \sin\left(\frac{x+y}{2}\right) = \sin\left(\frac{x-y}{2}\right)$ is

(1) $\ell n\left(\cot\frac{y}{2}\right) + 2\sin x = C$ (2) $\ell n\left(\cot\frac{y}{4}\right) + 2\sin\frac{x}{2} = C$

(3) $\ell n\left(\tan\frac{y}{2}\right) + 2\sin x = C$ (4) $\ell n\left(\tan\frac{y}{4}\right) + 2\sin\frac{x}{2} = C$

**MDE106**

**15.** The solution of the differential equation $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ is -

(1) $\ell n\left|\tan\left(\frac{x+y}{2}\right)+1\right| = x + c$ (2) $\ell n(\tan x) = y + c$

(3) $\ell n\tan\left(\frac{x+y}{2}\right) = x + c$ (4) $\ell n(\cos x) = y + c$

**MDE107**

**16.** The solution of differential equation $\frac{dy}{dx} = \frac{4x+6y+5}{3y+2x+4}$ is

(1) $x^2 + y^2 - xy + x - y = c$ (2) $y - 2x + \frac{3}{8}\ell n(24y + 16x + 23) = c$

(3) $4xy + 3(x^2 + y^2) - 10(x + y) = c$ (4) $y + 2x + \frac{3}{8}\ell n(24y + 16x + 23) = c$

**MDE108**

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**17.** The equation of the curve satisfying $\frac{dy}{dx} = \frac{y^2 - 2xy - x^2}{y^2 + 2xy - x^2}$ and passing through (1, –1) is .

(1) $x = y + c$ (2) $y = 2x + c$ (3) $y = x$ (4) $y = -x$

**MDE109**

**18.** The solution of differential equation $\left[x\cos\left(\frac{y}{x}\right) + y\sin\left(\frac{y}{x}\right)\right]y - \left[y\sin\left(\frac{y}{x}\right) - x\cos\left(\frac{y}{x}\right)\right]x\frac{dy}{dx} = 0$ is

(1) $y\cos\left(\frac{y}{x}\right) = c$ (2) $xy\sin\left(\frac{y}{x}\right) = c$ (3) $xy\cos\left(\frac{y}{x}\right) = c$ (4) $xy\tan\left(\frac{y}{x}\right) = c$

**MDE110**

**19.** The solution of the equation $\frac{dv}{dt} + \frac{k}{m}v = -g$ is

(1) $v = ce^{-\frac{k}{m}t} - \frac{mg}{k}$ (2) $v = c - \frac{mg}{k}e^{-\frac{k}{m}t}$ (3) $ve^{-\frac{k}{m}t} = c - \frac{mg}{k}$ (4) $ve^{\frac{k}{m}t} = c - \frac{mg}{k}$

**MDE111**

**20.** The solution of the differential equation $\frac{dy}{dx} = y\tan x - 2\sin x$ is

(1) $y = \cos x + c\sec x$ (2) $y = \cos x + c$

(3) $y = \sin x + c$ (4) $y = \operatorname{cosec} x + c$

**MDE112**



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**SECTION-B**

- This section will have **TEN** questions. Candidate can choose to attempt any 5 question out of these 10 questions. In case if candidate attempts more than 5 questions, first 5 attempted questions will be considered for marking.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value (Answer should be rounded off to the nearest integer).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4, if only correct answer is given.
Zero Marks : 0, if no answer is given.
Negative Marks : -1 for incorrect answer

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**1.** If $y(x)$ is the solution of the differential equation $(x+2)\frac{dy}{dx} = x^2 + 4x - 9$, $x \neq -2$ and $y(0) = 0$, then $y(-4)$ is equal to:

**MDE113**

**2.** The value of $\lim\limits_{x\to\infty} y(x)$ obtained from the differential equation $\frac{dy}{dx} = y - y^2$, where $y(0) = 2$ is

**MDE114**

**3.** Consider the differential equation, $ydx - (x + y^2)dy = 0$. If for $y = 1, x$ takes value 1, then value of $x$ when $y = 4$ is :

**MDE115**

**4.** If $\frac{dy}{dx} + y\tan x = \sin 2x$ and $y(0) = 1$, then $|y(\pi)|$ is equal to :

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**MDE116**

**5.** The solution of the differential equation $ydx - (x + 2y^2)dy = 0$ is $x = f(y)$. If $f(-1) = 1$, then $f(1)$ is equal to :

**MDE117**

**6.** If the solution of $\frac{dy}{dx} = \frac{ax+3}{2y+1}$ represents a circle, then the value of $|a|$ is

**MDE118**

**7.** Let $y$ be an implicit function of $x$ defined by $x^{2x} - 2x^x \cot y - 1 = 0$. Then $|y'(1)|$ equals :-

**MDE119**

**8.** If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, then $y(\ln 2)$ is equal to :-

**MDE120**

**9.** If $y(x)$ is a solution of the differential equation $\frac{dy}{dx} + 3y = 2$, then $\lim\limits_{x\to\infty} y(x)$ is equal to $\frac{k}{3}$ then value of $k$-

**MDE121**

**10.** If differential equation of family of curves $y\ell n|cx| = x$, where $c$ is an arbitrary constant, is $y' = \frac{y}{x} + \phi\left(\frac{x}{y}\right)$, for some function $\phi$, then $\frac{1}{|\phi(2)|}$ is equal to :

**MDE122**



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# JEE (Advanced) Practice Paper

This paper is for yourself practice and assessment the discussion of this paper is optional though you can see PDF solutions or video solutions or solutions in hardcopy whichever is provided.

## SECTION–I

- This section contains **SIX** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3, if only the bubble corresponding to the correct option is darkened.
Zero Marks : 0, if none of the bubbles is darkened.
Negative Marks : –1, in all other cases

**1.** The solution of the differential equation $(x^2 \sin^3 y - y^2 \cos x)dx + (x^3 \cos y \sin^2 y - 2y \sin x)dy = 0$ is

(A) $x^3 \sin^3 y = 3y^2 \sin x + C$ (B) $x^3 \sin^3 y + 3y^2 \sin x = C$
(C) $x^2 \sin^3 y + y^3 \sin x = C$ (D) $2x^2 \sin y + y^2 \sin x = C$

**MDE123**

**2.** Solution of differential equation $xy(my\,dx + nx\,dy) = \frac{xdy + ydx}{x^m y^n}$, given $m + n = 1$, is

(A) $x^{m+1}.y^{n+1} + 1 = c(x/y)$ (B) $x^{m+1}.y^{n+1} + 1 = cxy$
(C) $x^{m+1}.y^{n+1} - 1 = cxy$ (D) $x^m.y^n + 1 = cxy$.

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**MDE124**

**3.** $f(x)$ is a continuous and differentiable function defined in $\in [0, \infty)$. If $f(0) = 1$ and $f'(x) > 3f(x)\ \forall\ x \geq 0$ then

(A) $f(x) \leq e^{3x}\ \forall\ x \geq 0$ (B) $f(x) \leq e^{-3x}\ \forall\ x \geq 0$
(C) $f(x) > e^{3x}\ \forall\ x \geq 0$ (D) $f(x) \geq e^{3x}\ \forall\ x \geq 0$

**MDE125**

**4.** Find the curve which passes through the point (2, 0) such that the segment of the tangent between the point of tangency & the $y$–axis has a constant length equal to 2.

(A) $y = \pm\left[\sqrt{4-x^2} - 2\ell n\frac{2-\sqrt{4-x^2}}{x}\right]$ (B) $y = \pm\left[\sqrt{4-x^2} + 2\ell n\frac{2-\sqrt{4-x^2}}{x}\right]$
(C) $y = \pm\left[\sqrt{4-x^2} + 2\ell n\frac{2+\sqrt{4-x^2}}{x}\right]$ (D) $y = \pm\left[\sqrt{4-x^2} - 2\ell n\frac{2+\sqrt{4-x^2}}{x}\right]$

**MDE126**

**5.** Given $y = f(x)$ is a solution of differential equation $xdy + xdx = ydy - ydx$, also $f(0) = 0$, $f(1) = \tan\frac{3\pi}{8}$, then the area bounded by $y = f(x)$ on $x$-axis & $x = 0$ to $x = \tan\frac{\pi}{8}$ is equal to

(A) $\frac{1}{2}(\sqrt{2}-1)$ (B) $\frac{1}{2}(\sqrt{2}+1)^3$ (C) $\frac{1}{2}(\sqrt{2}-1)^3$ (D) $\frac{1}{2}(\sqrt{2}+1)^2$

**MDE127**



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**6.** Solution of differential equation $\frac{dy}{dx}=\frac{1}{xy(x^2\sin y^2+1)}$ is

(where $C$ is integral constant)

(A) $e^{y^2}\left(\frac{1}{x^2}-\frac{\cos y^2}{2}+\frac{\sin y^2}{2}\right)=C$

(B) $e^{y^2}\left(\frac{1}{x^2}+\frac{\cos y^2}{2}-\frac{\sin y^2}{2}\right)=C$

(C) $e^{y^2}\left(\frac{1}{x^2}-\frac{\cos y^2}{2}+\frac{\sin^2 y}{2}\right)=C$

(D) $e^{y^2}\left(\frac{1}{x^2}-\frac{\cos y}{2}+\frac{\sin y}{2}\right)=C$

**MDE128**

## SECTION-II

- This section contains **FIVE** questions.
- Each question has **FOUR** options for correct answer(*s*). **ONE OR MORE THAN ONE** of these four option(*s*) is (are) correct option(*s*).
- For each question, choose the correct option(*s*) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| *Full Marks* | : | +4 | if only (all) the correct option(*s*) is (are) chosen. |
| *Partial Marks* | : | +3 | if all the four options are correct but ONLY three options are chosen. |
| *Partial Marks* | : | +2 | if three or more options are correct but ONLY two options are chosen, both of which are correct options. |
| *Partial Marks* | : | +1 | if two or more options are correct but ONLY one option is chosen and it is a correct option. |
| *Zero Marks* | : | 0 | if none of the options is chosen (i.e. the question is unanswered). |
| *Negative Marks* | : | –2 | in all other cases. |

**For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(*s*) (second option in this case), with or without selection of any correct option(*s*) will result in –2 marks.

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**7.** A function $f : R \to R$ satisfies the differential equation $2xy + (1 + x^2)y' = 1$, where $f(0) = 0$, then

(A) $f(x)$ is odd

(B) $f(x) = 1$ has no real solution

(C) $f(x)=\frac{1}{2}$ has exactly two distinct solution

(D) $f(x) = -1$ has no real solution

**MDE129**

**8.** A conic passing through (2,2), has the property that slope of the tangent at any point $P$, on the curve is equal to reciprocal of the ordinate of point $P$ then -

(A) focus of the conic is $\left(\frac{1}{2},0\right)$

(B) latus rectum is equal to a prime number

(C) it passes through (8, –4)

(D) $\lim_{x\to\infty}\frac{y^2}{x}=2$, where $(x, y)$ lies on the curve

**MDE130**



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**9.** Let graph of $f(x)$ passes through $(0, 2)$ and tangent at any point $(x, y)$ of $f(x)$ intersects $x$-axis at $(x + 2, 0)$, then -

(A) $\lim\limits_{x\to\infty} f(x)=0$

(B) $\lim\limits_{x\to\infty} f(x)=2$

(C) equation of normal at $(0, 2)$ is $x - y + 2 = 0$

(D) equation of normal at $(0, 2)$ is $x + y - 2 = 0$

**MDE131**

**10.** Solution of differential equation $f(x)\dfrac{dy}{dx} = f^2(x) + f(x)\,y + f'(x)y$ is

(A) $y = f(x) + ce^x$ (B) $y = -f(x) + ce^x$ (C) $y = -f(x) + ce^x f(x)$ (D) $y = cf(x) + e^x$

**MDE132**

**11.** A differentiable function satisfies equation $f(x) = \int_0^x \left(f(t)\cos t - \cos(t-x)\right)dt$ then

(A) $f''\left(\dfrac{\pi}{2}\right) = e$ (B) $\lim\limits_{x\to-\infty} f(x) = 1$

(C) $f(x)$ has minimum value $1 - e^{-1}$ (D) $f'(0) = -1$

**MDE133**

## SECTION–III

- This section contains **ONE** paragraph.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 if only the bubble corresponding to the correct answer is darkened.
*Zero Marks* : 0 in all other cases.

**Comprehension # 1 (Q. No. 12 - 13)**

Consider a curve $C$ passing through the point $(1, 1)$ and has the property that the perpendicular distance of the normal at any point $P$ on the curve from origin is equal to the perpendicular distance of $P$ from $x$-axis.

On the basis of above information, answer the following questions :

**12.** Area of the smaller region bounded by curve '$C$' and $x^2 - 3y^2 - 2x + 1 = 0$

(A) $\dfrac{\pi}{2}$ (B) $\dfrac{\pi}{4}$ (C) $\dfrac{\pi}{6}$ (D) $\dfrac{\pi}{8}$

**MDE134**

**13.** Maximum distance of curve '$C$' from any point in set $A = \{(x, y) : y^2 + 4x - 4 = 0; x \in [0, 1]\}$

(A) $\sqrt{5}+1$ (B) $2\sqrt{2}$ (C) $\sqrt{5}-1$ (D) $2\sqrt{2}-1$

**MDE135**



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## SECTION–IV

- This section contains **TWO** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 If only correct answer is given.
*Zero Marks* : 0 If no answer is given.
*Negative Marks* : –1 For incorrect answer

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**14.** If $y = f(x)$ is solution of the differential equation $\frac{dy}{dx} - y = -e^x \tan^2 x$; $y(0) = 0$ and $g(x) = e^x$, $h(x) = e^x(1 - \sec^2 x)$.

Match List-I with List-II and select the correct answer using the code given below the list.

| | **List-I** | | **List-II** |
|---|---|---|---|
| (P) | The number of points of inflection of $f(x)$ in $\left(0, \frac{\pi}{2}\right)$ is (are) | (1) | 3 |
| (Q) | If $\left\lvert\lim_{x\to 0} \frac{f(x)}{x^3 g(x)}\right\rvert = \frac{a}{b}$, where $a$ and $b$ are coprime, then $(b - a)$ is equal to | (2) | 0 |
| (R) | If $\int_0^{\pi} (f(x) + h(x))\,dx = ke^{\pi}$, then $[k]$ is (where [.] denotes greatest integer function) | (3) | 2 |
| (S) | The number of solution(s) of the equation $f(x) = 0$ in $[0, \pi)$ is (are) | (4) | 1 |

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**Codes :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 1 | 3 | 4 | 1 |
| (B) | 3 | 3 | 4 | 1 |
| (C) | 2 | 3 | 1 | 4 |
| (D) | 3 | 1 | 2 | 4 |

**MDE136**



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**15.** Match List-I with List-II and select the correct answer using the code given below the list.

| | **List-I** | | **List-II** |
|---|---|---|---|
| (P) | Solution of differential equation $y(2x^4+y)\frac{dy}{dx}=(1-4xy^2)x^2$ is | (1) | $y^2=(\ell nx)^2+C$ |
| (Q) | Solution of differential equation $\frac{\sum_{n=1}^{\infty}\frac{x^{2n-1}}{(2n-1)!}}{\sum_{n=1}^{\infty}\frac{x^{2n-2}}{(2n-2)!}}=\frac{dx-dy}{dx+dy}, n\in N$ is | (2) | $3(x^2y)^2+y^3-x^3=C$ |
| (R) | Solution of differential equation $x=\sum_{r=0}^{\infty}\frac{(xy)^r}{r!}\left(\frac{dy}{dx}\right)^r$ is | (3) | $e^{x^2}\left(y^2-1\right)+e^{y^2}=C$ |
| (S) | Solution of differential equation $\left(e^{x^2}+e^{y^2}\right)y\frac{dy}{dx}+e^{x^2}\left(xy^2-x\right)=0$ is | (4) | $2ye^{2x}=ce^{2x}-1$ |

**Codes :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 2 | 1 | 4 | 3 |
| (B) | 1 | 4 | 2 | 3 |
| (C) | 2 | 4 | 1 | 3 |
| (D) | 3 | 2 | 4 | 1 |

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**MDE137**

## SECTION-V

- This section contains **THREE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9, **BOTH INCLUSIVE**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| *Full Marks* | : | +3 | If ONLY the correct integer is entered; |
| *Zero Marks* | : | 0 | If the question is unanswered; |
| *Negative Marks* | : | –1 | In all other cases. |

**16.** If $y(x)$ satisfies the differential equation; $\cos^2x\,(dy/dx) - (\tan 2x)\,y = \cos^4x$, $|x|<\frac{\pi}{4}$, and $y(0) = 0$ then $\frac{64y\left(\frac{\pi}{6}\right)}{3\sqrt{3}}$ is equal to

**MDE138**



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**17.** If $y = f(x)$ be a curve passing through $(e, e^e)$ and which satisfy the differential equation $(2ny + xy\log x)dx - x\log x\, dy = 0$, value of $\int_{1/e}^{e} g(x)\,dx$ where $g(x) = \lim_{n\to\infty} f(x)$, is :

**MDE139**

**18.** If $y = f(x)$ $(y > 0)$ satisfies the differential equation $x\frac{dy}{dx} - y = \frac{x^4}{y}$ such that $y(1) = 2$, then the value of $f(-1)$ is

**MDE140**

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# ANSWER KEY

## EXERCISE - O

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | A | A | A | B | C | B | A | B | D |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | A | C | A,B,D | A,B,D | A,C,D | A,C,D | A,B | A,B,D | A,B,D | B |
| Que. | 21 | 22 | 23 | | | | | | | |
| Ans. | A | A | B | | | | | | | |

## EXERCISE - S

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** | 49 | **2.** | 4 | **3.** | 2 | **4.** | 2 | **5.** | 2 |
| **6.** | 1 | **7.** | 1 | **8.** | 4 | **9.** | 2 | **10.** | 4 |

## EXERCISE - JEE (Main) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 1 | 1 | 1 | 2 | 2 | 1 | 2 | 2 | 3 | 3 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | | | |
| Ans. | 2 | 4 | 1 | 4 | 2 | 12 | 1 | | | |

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## EXERCISE - JEE (Advanced) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | D | A | C | D | B | A,C | B,C | A,D | A | B |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | A,C | B,C | B,C,D | 0.4 | A,D | A,C | A,C | 8 | C | C |
| Que. | 21 | 22 | | | | | | | | |
| Ans. | A,B,C | 16 | | | | | | | | |

## JEE (Main) Practice Paper

| | Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Section-A | A. | 2 | 4 | 3 | 4 | 2 | 4 | 3 | 4 | 2 | 2 |
| | Q. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| | A. | 3 | 3 | 1 | 4 | 1 | 2 | 4 | 3 | 1 | 1 |
| Section-B | Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| | A. | 0 | 1 | 16 | 5 | 3 | 2 | 1 | 7 | 2 | 4 |



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## JEE (Advanced) Practice Paper

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Section-I** | **Q.** | **1** | **2** | **3** | **4** | **5** | **6** | |
| | **A.** | A | B | D | B | A | A | |
| **Section-II** | **Q.** | **7** | **8** | **9** | **10** | **11** | | |
| | **A.** | A,B,D | A,B,C,D | A,C | C | A,D | | |
| **Section-III** | **Q.** | **12** | **13** | | | | | |
| | **A.** | C | A | | | | | |
| **Section-IV** | **Q.** | **14** | **15** | | | | | |
| | **A.** | C | C | | | | | |
| **Section-V** | **Q.** | **16** | **17** | **18** | | | | |
| | **A.** | 8 | 0 | 2 | | | | |

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# Important Notes

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